

Referential Alignment for Infant-Scale Vision-Language Models

Cross-situational grounding as balanced optimal transport

A research proposal. Scope: the idea, its motivation, worked examples, current evidence and known limitations.

Summary

A child learns which word means which thing from about 100 million words of input. Vision-language models trained at that scale do not: a recent from-scratch model reaches **32.4%** on a word-to-picture task against a **25%** chance floor. We argue the cause is structural — the training objective contains **no variable representing “this word refers to that region”** — and propose adding one. Each scene becomes a small assignment problem between the content words heard and the regions seen, solved under two constraints: a word may refer to **nothing**, and no region may absorb **everything**. The second constraint is mutual exclusivity, a documented bias in toddlers, and it turns out to be exactly a capacity constraint on an assignment matrix. Solving it costs about eight matrix multiplications.

1. The problem

Quine imagined a linguist who hears a native speaker say “**gavagai**” as a rabbit runs past. Does it mean rabbit, white, running, or dinner? One scene cannot say. An infant faces this continuously and solves it; a model trained on comparable data does not.

Two properties of transcribed caregiver speech make the problem harder than the textbook version, and both are visible in ordinary sentences:

utterance	content words (rows)	what should happen
look at the ball	ball	→ the region containing the ball
are you hungry	hungry	→ NOTHING (nothing on screen is “hungry”)
oh look are you hungry there is the ball	hungry, ball	hungry → NOTHING ball → the ball region
where did the ball go	ball	→ NOTHING (the ball has just left the frame)
put the cup on the table	put, cup, table	cup, table → their regions put → NOTHING (an action, not a thing)
there we go okay come on	come	→ NOTHING (all of it is filler)

Most word tokens refer to nothing visible. In four of the six sentences above, every content word should be left unattached. **And a few objects are in almost every frame and are almost never named** — hands, floor, table, the caregiver’s own body. They are permanently available to be blamed for any word.

Developmental psychology names the two mechanisms infants use. **Cross-situational learning**: no single scene resolves “ball”, but across hundreds of scenes the word co-occurs with balls and with nothing else. **Mutual exclusivity**: shown a familiar ball and an unfamiliar gadget and asked to “find the dax”, a toddler picks the gadget. Both are design targets for the objective in §3.

2. Why the current objective cannot fix it

Contemporary infant-scale vision-language models are trained by next-token prediction: given the image, predict the next word. That objective can be minimised from global scene gist plus the statistics of language — “are you...” is usually followed by “hungry” — **without ever binding a word to a thing**. There is no term in the loss that creates pressure to.

2.1 Why the obvious repair makes matters worse

The natural fix is to add a word-to-region assignment, computed the standard way: a row-wise softmax in which each word independently picks its best-matching region. In controlled simulation under realistic conditions, this is **worse than adding nothing at all**.

Objective	word-to-picture accuracy
captioning only	0.392
+ naïve alignment (row-softmax, every word must attach)	0.000
+ the two constraints of §3	0.700

Zero, with zero variance across three seeds, below the 0.025 chance floor. The reason is the first table: if roughly 80% of heard words refer to nothing visible and the objective **forces** each of them onto some region, then “hungry”, “come” and “put” are all attached to whatever happens to be on screen — usually a hand or the floor. Repeated across a corpus, the entire vocabulary collapses onto a handful of ever-present regions.

3. The proposal

Add the missing variable, then constrain it. Each scene is treated as an assignment problem. The **rows** are the content words just heard. The **columns** are regions of the image — not words — plus one extra column labelled **NOTHING**. Every word must be assigned somewhere.

	hand	floor	ball	table	NOTHING
ball			■		
hungry					■

Figure 1. “oh look — are you hungry? there’s the ball!” as an assignment matrix. Function words are filtered out; “ball” claims a region and “hungry” claims the NOTHING column.

Rule 1 — a word may refer to nothing

This is the NOTHING column. It sounds like a technicality and is the single most important element: it is what removes the forcing that produced the 0.000 above. The model is permitted to decline to ground a word, and must learn when to.

Rule 2 — no region may absorb everything

Each column carries limited capacity. If a region has already been claimed, the next word must look elsewhere. This is **mutual exclusivity** expressed as a constraint rather than a heuristic:

the degenerate solution in which every word points at one ever-present region becomes **infeasible**, not merely disfavoured.

Solving it, and remembering across scenes

“Assign rows to columns under capacity limits” is a classical problem — **optimal transport** — and the standard algorithm alternately normalises rows and columns: roughly eight matrix multiplications, negligible beside the vision encoder. A single scene is still hopeless, so a running tally records how often each word has been assigned to each kind of region, and feeds back into the next scene. The tally is scored by **how surprising** a pairing is rather than how frequent — raw counts would conclude that every word goes with hands, since hands are always present. That correction is what lets “ball” stand out, and it is the cross-situational part of the proposal.

4. Why we think this is the right shape of solution

Property	Why it matters
One parameter separates it from the standard method	A single scalar controls the column constraint. At zero the mathematics reduces exactly to the row-wise softmax that region-word contrastive methods already use. Every ablation is therefore the same code path with one number changed, so no baseline can be disadvantaged by a re-implementation.
A cognitive bias becomes a constraint	Mutual exclusivity has been studied for four decades as a behavioural phenomenon. Here it is precisely a capacity constraint on an assignment matrix, and enforcing it is nearly free.
Non-reference is represented, not assumed away	Existing objectives force every word onto some region. Given that most caregiver speech is non-referential, this is a defect rather than a simplification.
Localisation comes out for free	Regions are fixed cells of the image, so a word’s assigned column maps back to a known area. No bounding-box supervision is used at any point.

5. Evidence so far

All results below are from a **controlled simulation** in which the true word-to-object mapping is known, so we can measure whether the model learned the correct mapping rather than merely scoring well on a downstream task. Three seeds throughout; readout is on held-out instances of each category.

Condition	clean	moderate	realistic
captioning only	0.392	0.242	0.058
+ naive alignment	0.650	0.358	0.000
+ NOTHING column only	0.658	0.358	0.017
+ NOTHING + exclusivity	0.700	0.400	0.075

- The captioning objective alone is the weakest condition in every regime.
- Sweeping the exclusivity parameter continuously gives an **interior optimum** (0.275 at zero, 0.383 at the optimum, 0.350 at full exclusivity), and the rate at which words collapse onto never-named background regions falls as accuracy rises. The effect and the mechanism it is attributed to move together.

- Fitting a single free parameter — how many word-object pairs a learner encodes per trial — reproduces the human accuracy pattern of Yu & Smith (2007) with RMSE **0.066** (model 0.98 / 0.73 / 0.52 against human 0.89 / 0.78 / 0.56).

What is not yet established

Every number above is **simulation**. Whether the effect transfers to real video is untested and is the principal risk in the proposal. The pipeline accepts real image-utterance corpora and the evaluation is written, so this is a matter of staging data and spending a few GPU-hours rather than of further design.

6. Known limitations

Three of these are visible directly in the example sentences of §1, which is why they are stated here rather than discovered later.

what goes wrong	example	consequence
adjectives and nouns compete	here is the big blue truck → big, blue, truck	Three rows for one object. Exclusivity pushes them onto different regions, which is wrong. The constraint is a claim about nouns competing for referents and the objective does not know the difference.
verbs are treated as candidate referents	put the cup on the table → put, cup, table	“put” becomes a row and must be routed to NOTHING by the model, since no filter marks it as an action.
filler leaks through the filter	there we go okay come on → come	The stop-list is closed-class and fixed; it cannot know that this “come” is not referential. The NOTHING column is what has to absorb it.
many regions, few words	3 words, 16 regions	Each word carries 1/N of the mass while a region may hold only a small share, so full exclusivity forces a word to spread across ≥ 11 regions. This is the mechanical reason the measured optimum is interior rather than at maximum.

7. Proposed next steps

- **Real-data transfer.** Train the same two conditions on a real image-utterance corpus at infant scale and evaluate word-to-picture and localisation accuracy. This is the experiment that decides whether the proposal is correct.
- **A referent-aware column marginal.** The many-regions/few-words tension above has a principled fix: make the capacity reflect how many regions plausibly contain objects, rather than spreading it uniformly.
- **Phrase-level rows.** Grouping “big blue truck” into one row before the assignment would remove the adjective conflict, at the cost of needing a chunker.
- **Estimating the non-reference rate from data.** The share of speech that refers to nothing is currently a hyper-parameter; it could be estimated from transcribed child-directed speech corpora.